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Commentary

Imaging and radiotherapy for recurrent prostate cancer: An evolutionary partnership

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We applaud the Australian and New Zealand Radiation Oncology Genito-Urinary Group for their exemplary review and recommendations of management for prostate cancer patients with recurrence [1]. Technological advances in imaging and radiotherapy have enabled 'new' disease states to be recognized and managed, however, these need to be robustly evaluated to ensure clinically appropriate endpoints are met before introducing into routine clinical care.

The manuscript amply illustrates the triumphal march of Positron Emission Tomography (PET) with radiolabeled ligands of the Prostate Specific Membrane Antigen (PSMA) during recent years. PSMA-ligand PET/CT may now play a role in all clinical states in prostate cancer.

PSMA is a transmembranous enzyme which is significantly overexpressed in most adenocarcinomas of the prostate gland, thereby presenting a promising target for imaging and endoradiotherapy. Following many years of outstanding preclinical research, the clinical breakthrough in PSMA-based PET-imaging was achieved with the invention of the PET tracer ^{68}Ga-PSMA-11 at the German Cancer Research Center (DKFZ Heidelberg) and its clinical introduction in May 2011 at the department of Nuclear Medicine of the Heidelberg University Hospital [2,3]. PSMA-11, also known by abbreviations such as HBED, HBED-CC or PSMA-HBED-CC, is a "small molecule" (low-molecular weight) ligand of PSMA with a very high receptor affinity. In contrast to antibodies which are of significantly bigger size, small and low-molecular weight molecules exhibit superior tissue penetration, distribution as well as rather rapid blood clearance. Antibodies binding to PSMA have also been developed, albeit without showing any convincing results regarding tumor visualization.

The initial as well as subsequent clinical results demonstrated that this novel method of imaging, commonly known as "PSMA-PET", is a significant step forward in the diagnosis of recurrent prostate cancer. Since then, ^{68}Ga-PSMA-11 PET/CT has rapidly spread worldwide. Meanwhile, various alternative and promising PSMA-ligands exist. However, the overwhelming majority of publications about PSMA-imaging are based on ^{68}Ga-PSMA-11.

Currently, only a few prospective publications including smaller patient cohorts exist. More prospective trials are eagerly awaited. Until then, a broader study of the literature beyond national publications can be of help in supporting or supplementing the recommendations made by the authors.

To date, clinical trials evaluating prostate cancer recurrence and management have not been prescriptive on imaging modality. The choice of imaging modality and results can affect the management of patients with primary or recurrent prostate cancer [4]. Hence, the need going forward is for clinical trials to incorporate novel imaging techniques into their protocols, potentially, as an adjunct to more traditional imaging modalities. This would then enable determination of whether PSMA-based management strategies result in improved patient outcomes, with chance of acceptance by the respective national health care systems.

Although PSMA-PET has maybe proven its worth as the most sensitive and specific diagnostic modality for the visualization of recurrent prostate cancer, tumor detection at very low PSA levels remains challenging. Publications exist which report low PET-positive rates of 11–23% as listed by the authors [1]. This is in contrast to other publications including bigger patient cohorts which report higher rates [5–7]. Also, it has to be mentioned that more aggressive prostate cancer exhibits less PSA production and higher PSMA expression [8] which possibly favors tumor detection in higher Gleason scores. Future prospective studies are mandatory to more accurately define precise estimates of sensitivity and specificity in particular clinical situations. Until then, the commentators suggest that the treating physicians should be left to make the decision as to whether a PSMA-PET/CT should be conducted in the context of a low PSA level. Patients may benefit from detection of an early cancer recurrence even if their PSA does not meet any pre-defined level.

The authors of the manuscript importantly apprehend the thought of not considering salvage radiotherapy in the absence of positive imaging findings. We need to safeguard from the idealism that negative PSMA scans simplify treatment decisions and should not be a sole factor in influencing management options for patients. In a retrospective series of patients with pathologic correlation, ^{68}Ga-PSMA-11 was false-negative in just over 20% of lymph node metastases [9].

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An important question in PSMA-imaging concerns the effect of androgen deprivation therapy (ADT). It is known from preclinical studies that ADT can increase the PSMA expression of prostate cancer cells [10,11]. However, this was observed exclusively for short-term ADT. Hope et al. demonstrated an increased tracer uptake in PSMA-PET/CT compared to base-line PET/CT in one patient four weeks after ADT was commenced [11]. However, it was uncertain whether or not the higher tracer uptake was not related to tumor progression. Recently, a study was published which demonstrated a significantly negative impact of long-term ADT on prostate cancer visibility in PSMA-PET/CT [12]. Therefore, the statement of the authors regarding ADT and PSMA-based imaging needs to be interpreted with great caution.

The recommendations of the Australian and New Zealand Radiation Oncology Genito-Urinary group have encompassed the evolving role of imaging which in part portrays the described clinically relevant scenarios we now face regularly in our multidisciplinary meeting. The major challenge is knowing which patients have potentially curable or life prolonging management options.

There have been rigorous reviews appraising timing of radiotherapy post-prostatectomy [13], but do we also need to be smarter regarding our target volumes? We eagerly await the results of the RTOG 0534 phase 3 study regarding the role of elective pelvic lymph node irradiation post-prostatectomy. There are at least four consensus guidelines for prostate bed contours, but when evaluated with the pre-operative prostate MR, none of them completely covered the pre-operative extent of the prostate [14]. Would the use of pre-operative MR and post-operative PSMA-PET/CT enable an individualized and targeted approach to post-operative prostate radiotherapy?

The Australian and New Zealand Radiation Oncology Genito-Urinary group highlight the awaited clinical trial results, which will aid to inform further guidance for these clinical settings. While clinical disease endpoints are the cornerstone for trials, we need to utilize prospective clinical trials to better appreciate the biology of these disease states. The ORIOLE study is an exemplar of this, aiming to evaluate the biology of the oligometastatic state using immunologic, cellular, molecular and functional imaging correlates [15]. Also, collaborative initiatives such as OligoCare, a European prospective registry trial of oligometastatic disease aspiring to generate robust data to help determine which patients are most likely to benefit from what treatment need to be encouraged.

PSMA-PET/CT informs our clinical management discussions and decisions with patients, but we need evidence from prospective clinical trials to justify its role and place in prostate cancer practice. We need to be aware of the risk of overtreatment in patients whose disease may never affect their overall survival or quality of life.

As George Bernard Shaw wisely stated, "Beware of false knowledge, it is more dangerous than ignorance".

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